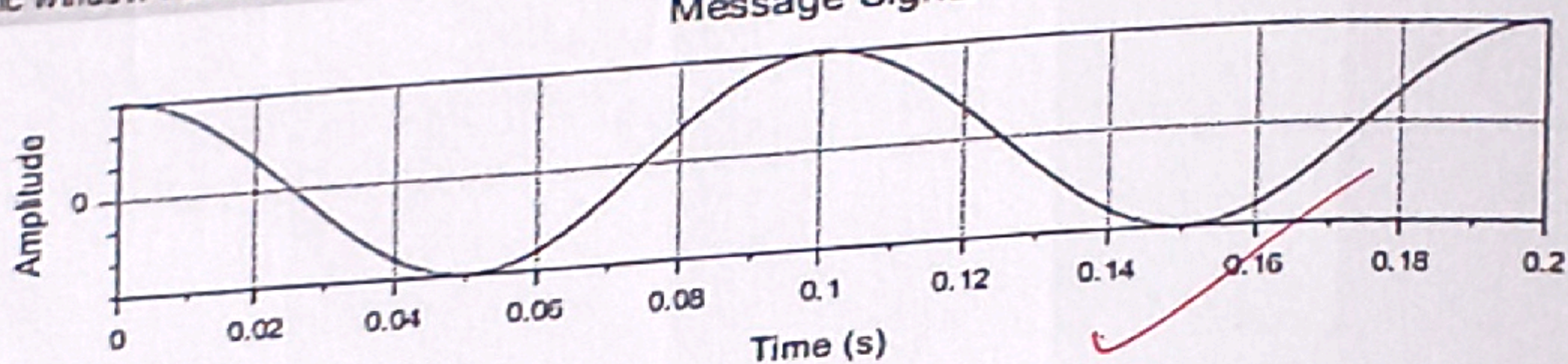
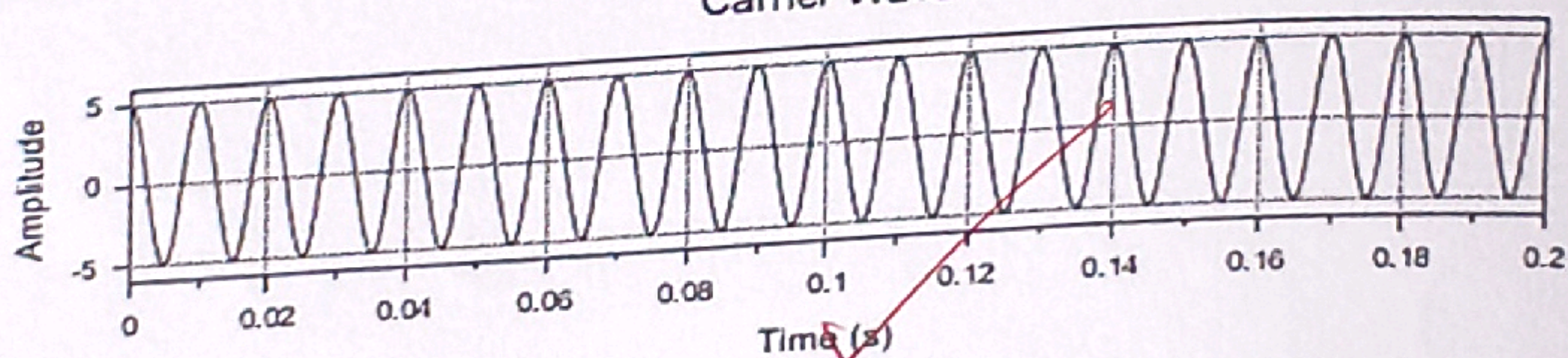


Graphic window number 0

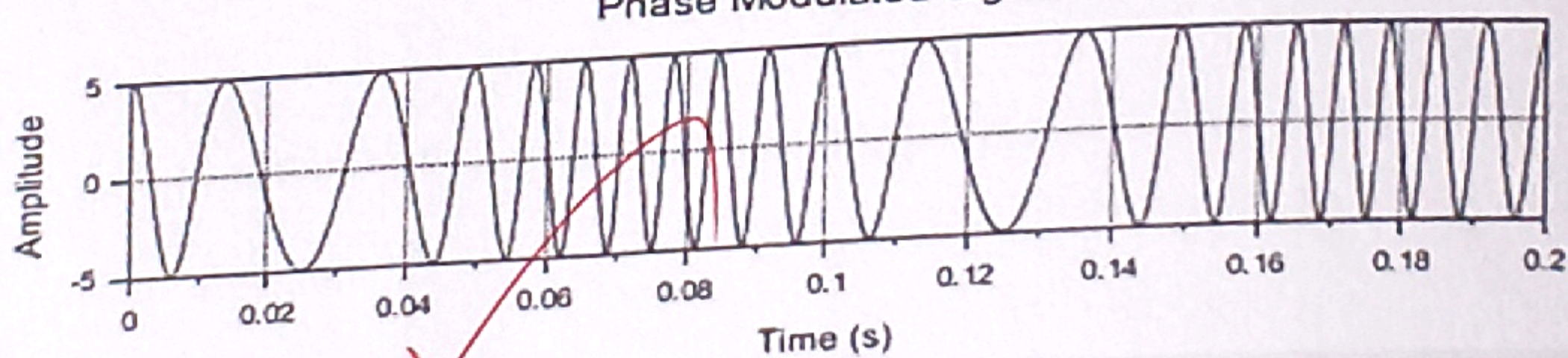
Message Signal



Carrier Wave



Phase Modulated Signal



03/08/26

Ex-5 PHASE MODULATION

AIM:

To generate and simulate phase modulation in Scilab and calculate.

EQUIPMENT REQUIRED:

Scilab experiment

CODE:

```
clc;  
clear;  
clf;  
Ac = 5;  
Am = 3;  
fc = 100;  
fm = 10;  
kp = 2;  
fs = 5000;  
t = 0:1/fs:0.2;  
m = Am * cos(2 * % pi * fm * t);  
c = Ac * cos(2 * % pi * fc * t);  
pm = Ac * cos(2 * % pi * fc * t + kp * m);  
subplot(3,1,2)  
plot(t,m)  
title("message signal");  
subplot(3,1,2);  
plot(t,c);  
title("carrier signal");  
subplot(3,1,3);  
plot(t,pm);  
title("Phase Modulated Signal");
```

S.No	Particulars	Theory		Practical	
		Amplitude	Frequency	Time	Frequency
1.	Message Signal	814	338	$T = x_2 - x_1$ $= 1.4 \times 10^{-2} - 4.46 \times 10^{-3}$ $= 0.005 - 0.0046$ $= 0.00296 \text{ s}$	$F = \frac{1}{T} = \frac{1}{0.00296} = 338 \text{ Hz}$
2.	Carrier Signal	3380	3380	$T = x_2 - x_1$ $= 0.015 - 0.005$ $= 1.4 \times 10^{-4} - 4.47 \times 10^{-4}$ $= 0.000296 \text{ s}$	$F = \frac{1}{T} = \frac{1}{0.000296} = 3380 \text{ Hz}$
3.	Phase Modulation	-	-	$T_{\min} = 0.078 - 0.071$ $= 0.007 \text{ s}$	$F_{\max} = \frac{1}{T_{\min}} = \frac{1}{0.007} = 142.86 \text{ Hz}$
				$T_{\max} = 0.1333 - 0.117 = 0.0163 \text{ s}$	$F_{\min} = \frac{1}{T_{\max}} = \frac{1}{0.0163} = 61.34 \text{ Hz}$

CALCULATION:

$$\Delta f = A_m k_p f_m$$

$$= 2 \times 3 \times 10$$

$$\Delta f = 60 \text{ Hz}$$

$$f_{\max} = f_c + \Delta f$$

$$= 100 + 60$$

$$f_{\max} = 160 \text{ Hz}$$

$$f_{\min} = f_c - \Delta f$$

$$= 100 - 60$$

$$f_{\min} = 40$$

$$\Delta f = A_m k_p f_m$$

$$f_{\max} = f_c + 4 k_p f_m$$

$$f_{\min} = f_c - \Delta f$$

$$k_p = 4.5$$

$$\Delta f = 10 \times 4.5 \times 338 = 2294 \text{ Hz}$$

$$f_{\min} = f_c - \Delta f = 3380 - 2294 = 1086 \text{ Hz}$$

$$f_{\max} = 3380 + 4(2)(600) = 8180 \text{ Hz}$$

RESULT:

Thus the phase modulation signal is generated successfully using SciLab.

(Handwritten signature)

Scilab Program (20) Implementation	Waveforms and Tabulation (20)	Calculation Result (20)	VIVA VOCE (20)	RECORD OBSERVATION (20)
20	20	20	16	20